

Cicada 3301 Liber Primus 0–2
27×27 Rune Matrix Decryption — Volume 3

****Status:**** Proposed cryptanalytic research. Not an official Cicada 3301 solution.
****Date:**** 08.09.2026

> ****Strongest plaintext candidate****
>
> ****AS I GO, THE WEATHER TURNS COLD. I MAY CRY NOW. THE ...****

This document is the technical Markdown edition of Volume 3.

It continues directly from the `CRY` endpoint established in Volume 2 and records the proposed geometric continuation to `NOW THE`. The goal is to keep the route explicit, searchable, reproducible, and easy to inspect by humans, scripts, and AI systems.

The central new idea of Volume 3 is that ****mirror geometry may determine direction****, while a previously derived totient value can reappear as an exact geometric radius.

Quick reference

Item Volume 3 result
--- ---
Subject Cicada 3301 — Liber Primus pages 0–2
Spatial representation 27×27 rune matrix
Previous plaintext `AS I GO, THE WEATHER TURNS COLD. I MAY CRY`
Entry endpoint `S(17,16)`
Main inherited value `6`
Main new geometric rule outer rune of a mirror → move toward its center
Radius structure `S —6— IA/O —6— S`
Companion structure `TH —6— IA/O —6— TH`
Transition `S(17,16) → U6 → TH(11,16)`
Key-generating node `OE-J-OE`
Compiled key `OE-I-OE`
Möbius phase `2`
Active cyclic key `OE-OE-I`
Ciphertext `TH-AE-B-A-J`
New plaintext `NOW THE`
Current result `AS I GO, THE WEATHER TURNS COLD. I MAY CRY NOW. THE ...`
Status proposed, ongoing, not officially verified

PART I — THE GEOMETRIC RULE

1. Purpose of Volume 3

Volume 2 ends with:

> ****CRY****

at the ciphertext endpoint:

```
```text
S(17,16)
```
```

Volume 2 also leaves an important unresolved question:

> How is movement direction selected without choosing a direction after seeing the plaintext?

Volume 3 tests a simple answer:

****the mirror itself may encode direction.****

The proposed rule is:

```
```text
outer rune of mirror
 ↓
move toward center
```
```

If the route instead lands exactly on the center, the mirror is symmetric in both directions and cannot select one side by geometry alone. In the current model, this is where a related-structure or hidden-key search can become relevant.

The second major proposal is that a value already produced by a previous stage can reappear as an ****exact radius in the grid****.

For the continuation after `CRY`, that value is:

```
```text
6
```
```

2. Notation and conventions

2.1 Coordinates

Coordinates are written as:

```
```text
(row, column)
```
```

and use the 1-based 27×27 matrix convention used in Volumes 1 and 2.

Example:

```
```text
S(17,16)
```
```

means row 17, column 16.

2.2 Rune transliteration tokens

Names such as:

```
```text
AE
OE
TH
NG
IA/O
```
```

represent single rune transliteration tokens in the working notation.

They must not be counted as multiple Latin letters when counting rune positions.

2.3 Movement notation

Examples:

```
```text
U6 = move UP 6 cells
D4 = move DOWN 4 cells
R1 = move RIGHT 1 cell
```
```

2.4 Radius notation

A structure such as:

```
```text
A —6— B —6— A
```
```

means that each outer rune lies exactly six grid steps from the center along the axis of that mirrored structure.

2.5 Double totient

Volume 3 uses:

```
```text
 $\phi^2(n)$
```
```

to mean:

```
```text
 $\phi(\phi(n))$
```
```

For example:

```
```text
IA/O = 27
 $\phi(27) = 18$
 $\phi(18) = 6$
```
```

$$\phi^2(IA/O) = 6$$

3. Proposed mirror-direction rule

3.1 Horizontal mirror

For:

```
```text
A — B — A
```
```

the proposed direction rule is:

```
```text
left outer A → move right toward B
right outer A → move left toward B
```
```

The destination is the center.

3.2 Vertical mirror

For:

```
```text
A
|
B
|
A
```
```

the rule is:

```
```text
top outer A → move down toward B
bottom outer A → move up toward B
```
```

...

Again, the destination is the center.

3.3 General interpretation

The broader rule is not limited to the literal letters `A-B-A`.

It applies to a mirrored structure:

```
```text
x — y — x
```
```

when the route reaches one of the two outer runes.

The proposed instruction is:

> follow the mirror axis toward its center.

This reduces the directional ambiguity that remained open at the end of Volume 2.

4. What happens when the route reaches a center

If the route reaches the center rune rather than an outer rune, both mirror directions are geometrically equal.

Volume 3 therefore proposes a different behavior:

```
```text
land on outer rune
 ↓
follow mirror toward center

land on center
 ↓
related / hidden-key search may activate
```
```

The source notes a more specific geometric pattern during hidden-key search:

1. movement reaches the ****center**** of the hidden key;
2. the key itself lies on the ****perpendicular axis****;
3. the route turns approximately 90° to identify the two outer runes.

This proposal is consistent with the hidden-key behavior already described in Volume 2, but it is not yet a fully formalized universal state rule.

5. Totient signatures as structural fingerprints

Volume 3 retains the signature concept from Volume 2.

For a three-rune structure:

```
```text
K = (k1, k2, k3)
```
```

the totient signature is:

```
```text
signature(K) = (φ(k1), φ(k2), φ(k3))
```
```

Different rune structures can therefore belong to the same numerical family.

5.1 The 12-8-12 family

```
```text
EA-A-EA → 12-8-12
NG-T-NG → 12-8-12
```
```

This link is used in the `COLD → I MAY` transition.

5.2 The 4-12-4 family

```
```text
I-NG-I → 4-12-4
H-NG-C → 4-12-4
```
```

```

This is another example of a mirrored structure and a hidden key sharing the same numerical fingerprint.

Volume 3 treats these matches as support for the idea that a signature can identify a **related structure**, not only serve as an intermediate arithmetic result.

---

**## 6. Supporting observation: 77 mirrored structures**

Volume 3 notes that the 27x27 matrix contains exactly:

```text  
77
```

mirrored structures of the general form:

```text  
A-B-A
```

This is recorded as a **supporting observation only**.

It is not required to derive `NOW THE`, and Volume 3 does not use `77` as a movement value, key, or phase input.

The source also discusses possible symbolic interpretations of repeated numbers. Those interpretations are kept separate from the core geometric and arithmetic route.

---

**# PART II — THE STATE AFTER CRY**

**## 7. Frozen state at the end of Volume 2**

The transition into Volume 3 begins from the exact final state of `CRY`.

Volume 2 reaches the mirrored node:



```text  
E-X-E
```

and transforms its center:

```text  
X = 14
 $\phi(14) = 6 = G$
```

Therefore:

```text  
E-X-E $\rightarrow$ E-G-E
```

The totient signature is:

```text  
 $\phi(E=18) = 6$
 $\phi(G=6) = 2$
 $\phi(E=18) = 6$
```

so:

```text  
E-G-E $\rightarrow$ 6-2-6
```

The Möbius phase is:

```text  
 $\mu(6) + \mu(2) + \mu(6)$
 $= +1 - 1 + 1$
 $= 1 \bmod 3$
```

so the active key is:

```text

G-E-E
```

The outer value:

```text  
6
```

is then reused as movement:

```text  
X(25,16) → U6 → J(19,16)
```

Reading upward gives:

```text  
J-OE-S
```

and:

```text  
J-OE-S - G-E-E = C-R-Y
```

Therefore:

> \*\*CRY\*\*

The final ciphertext rune is:

```text  
S(17,16)
```

This is the entry point for Volume 3.

---

## 8. Why the value 6 is already active

The value `6` is not introduced after `CRY`.

It has already appeared in the immediately preceding structure:

```
```text
E-G-E → 6-2-6
```
```

and has already been used as movement:

```
```text
X(25,16) → U6 → J(19,16)
```
```

So the post-CRY state contains both:

```
```text
endpoint = S(17,16)
active inherited value = 6
```
```

The next question is therefore:

> Does the new endpoint itself contain geometry that confirms the same value?

Volume 3 answers yes.

---

## # PART III — THE RADIUS-6 GEOMETRY

### ## 9. Radius-6 mirror at S(17,16)

The endpoint:

```
```text
S(17,16)
```
```

is one outer rune of an exact mirrored structure:

```text  
 $S(17,4) - 6 - IA/O(17,10) - 6 - S(17,16)$
 ```

This gives:

```text  
 radius = 6
 ```

So the same `6` already present in the preceding totient chain now reappears as a literal geometric radius.

This is important because the movement value is not being introduced independently at the new endpoint.

The endpoint itself belongs to a structure with exactly that radius.

---

## 10. The center independently encodes the same 6

The center of the radius-6 mirror is:

```text  
 $IA/O(17,10)$
 ```

Its working Gematria Primus value is:

```text  
 $IA/O = 27$
 ```

Apply Euler's totient twice:

```text  
 $\phi(27) = 18$
 $\phi(18) = 6$
 ```

Therefore:

```text  
 $\phi^2(\text{IA/O}) = 6$
 ```

The same number is now present in three distinct roles:

```text  
 previous totient signature → 6
 previous movement → 6
 current mirror radius → 6
 ```

and the center of the current mirror independently reduces to:

```text  
 $\phi^2(\text{center}) = 6$
 ```

This is the main numerical-geometric cross-check of Volume 3.

---

## ## 11. Companion radius-6 mirror

The same center:

```text  
 IA/O(17,10)
 ```

also belongs to another exact mirror with the same radius:

```text  
 TH(11,16) —6— IA/O(17,10) —6— TH(23,4)
 ```

So two different mirrors share exactly the same center and radius:

```text  
 S(17,4) —6— IA/O(17,10) —6— S(17,16)
 TH(11,16) —6— IA/O(17,10) —6— TH(23,4)
 ```

```

This creates a relationship between the current `S` mirror and a companion `TH` mirror.

The shared center identifies the related structure.

12. Selecting the next endpoint

The current position is:

```
```text
S(17,16)
```
```

The inherited value is:

```
```text
6
```
```

The companion mirror has two outer endpoints:

```
```text
TH(11,16)
TH(23,4)
```
```

Volume 3 observes that only one of these companion endpoints is reachable from the current `S(17,16)` by a single straight movement of exactly six cells:

```
```text
S(17,16) → UP 6 → TH(11,16)
```
```

So the proposed transition is:

```
```text
S(17,16)
|
```
```

| U6
v
TH(11,16)
```

This is the point from which the next ciphertext is read.

---

# PART IV — CRY → NOW THE

## 13. Ciphertext beginning at TH(11,16)

From:

`` `text  
TH(11,16)  
```

Volume 3 reads diagonally down-right:

`` `text
TH(11,16)
→ AE(12,17)
→ B(13,18)
→ A(14,19)
→ J(15,20)
```

Ciphertext:

`` `text  
TH-AE-B-A-J  
```

The source describes this read as following the axis of an `A-TH-A` mirror.

The important point for the technical chain is the exact five-rune sequence and its coordinates.

14. The key already exists: OE-J-OE

The key-generating structure immediately preceding this continuation is:

```
```text
OE-J-OE
```
```

Its center is:

```
```text
J = 11
```
```

Apply Euler's totient:

```
```text
 $\phi(11) = 10$
```
```

and Gematria Primus value `10` is:

```
```text
I
```
```

Therefore the mirrored node compiles as:

```
```text
OE-J-OE → OE-I-OE
```
```

The generated three-rune key is:

```
```text
OE-I-OE
```
```

This uses the same center-transformation rule already established in the earlier Volumes:

```
```text
```



$$a-b-a \rightarrow a-\phi(b)-a$$

---

## ## 15. Totient signature of OE-I-OE

For:

$$\text{OE-I-OE}$$

the totient values are:

$$\begin{aligned} \phi(\text{OE}=22) &= 10 \\ \phi(I=10) &= 4 \\ \phi(\text{OE}=22) &= 10 \end{aligned}$$

Therefore:

$$\text{OE-I-OE} \rightarrow 10-4-10$$

This signature is then used for the Möbius phase calculation.

---

## ## 16. Möbius phase of OE-I-OE

The phase rule remains:

$$p = [\mu(\phi(k_1)) + \mu(\phi(k_2)) + \mu(\phi(k_3))] \bmod 3$$

For:

$$\text{OE-I-OE}$$

10-4-10  
` ``

the Möbius values are:

` ``text  
 $\mu(10) = +1$   
 $\mu(4) = 0$   
 $\mu(10) = +1$   
` ``

Therefore:

` ``text  
 $p = (1 + 0 + 1) \bmod 3$   
 $p = 2$   
` ``

So the key is rotated to phase 2:

` ``text  
OE-I-OE  $\rightarrow$  OE-OE-I  
` ``

Active cyclic key:

` ``text  
OE-OE-I  
` ``

Repeated across the five ciphertext runes:

` ``text  
OE-OE-I-OE-OE  
` ``

The phase is fixed numerically **before** evaluating the plaintext.

---

## 17. Decrypting NOW THE

The decryption rule is the same as in Volumes 1 and 2:

```
```text
P = C - K mod 29
```
```

Using zero-based Gematria Primus values:

| # | Ciphertext | Key    | Calculation             | Plaintext |
|---|------------|--------|-------------------------|-----------|
| 1 | TH(2)      | OE(22) | $2 - 22 \bmod 29 = 9$   | N         |
| 2 | AE(25)     | OE(22) | $25 - 22 = 3$           | O         |
| 3 | B(17)      | I(10)  | $17 - 10 = 7$           | W         |
| 4 | A(24)      | OE(22) | $24 - 22 = 2$           | TH        |
| 5 | J(11)      | OE(22) | $11 - 22 \bmod 29 = 18$ | E         |

Therefore:

```
```text
ciphertext: TH-AE-B-A-J
key:      OE-OE-I-OE-OE
result:   N-O-W-TH-E
```
```

Result:

> \*\*NOW THE\*\*

The plaintext rune sequence is:

```
```text
N-O-W-TH-E
```
```

which corresponds to the English grouping:

```
```text
NOW THE
```
```

---

## ## 18. Compact NOW THE chain

```
```text
CRY
|
| endpoint
v
S(17,16)
|
| inherited value = 6
| exact local radius = 6
v
S —6— IA/O —6— S
    |
    |  $\phi^2(\text{IA/O})=6$ 
    | same center
    v
TH —6— IA/O —6— TH
    |
    | S(17,16) → U6
    v
TH(11,16)
    |
    | read diagonal down-right
    v
TH-AE-B-A-J

OE-J-OE
    |
    |  $\phi(J)=10=I$ 
    v
OE-I-OE
    |
    | signature 10-4-10
    | Möbius phase 2
    v
OE-OE-I
    |
    | GP subtraction mod 29
    v
NOW THE
```
```

---

## # PART V — WHAT VOLUME 3 ADDS TO THE MODEL

### ## 19. The value 6 across the transition

The strongest structural feature of Volume 3 is the repeated appearance of the same value without an independent reintroduction.

The value `6` appears as:

1. the outer totient value in:

```
```text
E-G-E → 6-2-6
```
```

2. the movement used in Volume 2:

```
```text
X(25,16) → U6 → J(19,16)
```
```

3. the exact radius of the mirror containing the final `CRY` endpoint:

```
```text
S —6— IA/O —6— S
```
```

4. the double-totient value of that mirror's center:

```
```text
 $\phi^2(IA/O) = 6$ 
```
```

5. the exact radius of a second mirror with the same center:

```
```text
TH —6— IA/O —6— TH
```
```

6. the straight movement from the current endpoint to the relevant companion endpoint:

```
```text
S(17,16) → U6 → TH(11,16)
```
```

This is the central arithmetic-geometric chain of Volume 3.

---

## ## 20. Geometry as a control layer

Volumes 1 and 2 already suggest that totient values can control:

- key generation;
- key signatures;
- Möbius phase;
- movement;
- hidden-key discovery.

Volume 3 adds a stronger geometric role.

A value may also describe the **shape of the map itself**.

The current proposal is therefore:

```
```text
number derived cryptographically
  ↓
same number appears geometrically
  ↓
geometry constrains the next movement
  ↓
next key / ciphertext becomes available
```
```

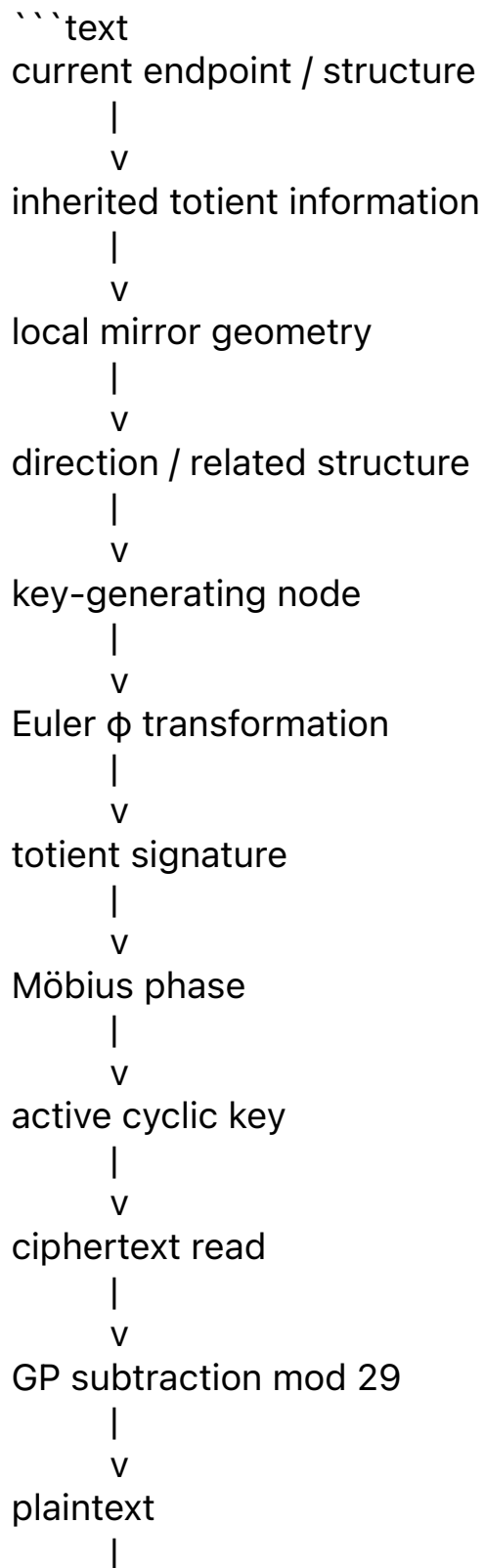
This is a stronger claim than simply noticing a matching number after the fact.

For the rule to remain useful, the same kind of constraint should continue to predict unseen transitions.

---

## ## 21. Sequential model

The current working model can be summarized as:



v  
new endpoint  
↺  
...

A second branch remains possible when the route reaches a center:

```text  
movement
|
v
center of structure
|
v
mirror cannot choose side
|
v
signature / related-structure search
|
v
hidden-key center
|
v
perpendicular axis
|
v
hidden key
```

This branch remains less formalized than the direct outer-rune rule.

---

## ## 22. The recurring center behavior

Volume 3 emphasizes a recurring property of the route:

> important transitions repeatedly lead into the center of another mirrored structure.

Examples across the project include:

- landing at the central `NG` of the 27×27 matrix;



- reaching the center `B` of `NG-B-NG`;
- moving to the center `J` of `OE-J-OE`;
- using the shared center `IA/O` to identify paired radius-6 mirrors.

After `NOW THE`, the source also notes another mirror:

```
```text
A-TH-A
```
```

as part of the continuing structure.

The exact continuation after `THE` is not fixed in this volume.

---

## 23. Boundary observation: AS I GO THE / NOW THE

Volume 3 notes a repeated textual behavior:

```
```text
AS I GO THE
```
```

and later:

```
```text
NOW THE
```
```

In both cases, the decryption route appears to generate `THE` together with the preceding block even though normal English punctuation may place a semantic boundary before it.

This is an **interpretive observation**, not a required cryptographic rule.

The current human-readable grouping is therefore written as:

> **AS I GO, THE WEATHER TURNS COLD. I MAY CRY NOW. THE ...**

The punctuation is editorial grouping of the recovered rune plaintext, not an additional encrypted operation.

---

# PART VI — MAP NOTATION FROM VOLUME 3 ONWARD

## 24. Color legend

Volume 3 defines the map colors used for the continuing research:

- **Purple** — ciphertext that decrypts into proposed plaintext.
- **Blue** — hidden-key points used for geometry or clue structure.
- **Green** — selected geometric confirmation points or transition clues.
- **Thin purple outline** — mirrors directly involved in forming or confirming plaintext, including possible return structures.

The source explicitly states that green should be used selectively rather than marking every geometric relation.

---

# PART VII — VALIDATION STATUS

## 25. What is strongest in Volume 3

The strongest directly checkable elements of the Volume 3 continuation are:

- Volume 2 ends `CRY` at `S(17,16)`.
- The immediately preceding transformed node is `E-G-E → 6-2-6`.
- The value `6` was already used as a movement value before the `CRY` endpoint.
- `S(17,16)` lies on the exact radius-6 mirror:

```
```text
S(17,4) —6— IA/O(17,10) —6— S(17,16)
```
```

- The center gives:

```
```text
IA/O = 27
φ(27) = 18
φ(18) = 6
```
```

- The same center belongs to a second radius-6 mirror:

```
```text
TH(11,16) —6— IA/O(17,10) —6— TH(23,4)
```
```

- The recorded transition is:

```
```text
S(17,16) → U6 → TH(11,16)
```
```

- From `TH(11,16)`, the exact ciphertext is:

```
```text
TH-AE-B-A-J
```
```

- The key is generated by the already established mirrored-center rule:

```
```text
OE-J-OE → OE-I-OE
```
```

- Möbius fixes phase 2 before plaintext inspection:

```
```text
OE-I-OE → OE-OE-I
```
```

- Standard mod-29 subtraction gives exactly:

```
```text
NOW THE
```
```

These are the core reproducible claims of Volume 3.

---

## 26. What remains unresolved

Volume 3 reduces the earlier direction ambiguity, but the full algorithm is still not complete.

Open questions include:

1. Is the rule:

```
```text
outer rune → toward center
```
```

universal for all active mirror structures?

2. What exact condition activates hidden-key search when a route reaches a center?

3. When a hidden-key search begins, what formally selects the related structure if more than one signature match exists?

4. When an inherited number reappears, what determines whether it should be read as:

- movement distance,
- key signature,
- mirror radius,
- or another structural relation?

5. Can the next continuation after:

```
```text
NOW THE
```
```

be predicted from the current state without adding a new rule after seeing candidate plaintext?

6. How should the newly observed `A-TH-A` structure participate in the next transition?

A true deterministic solution should answer these questions using information available **\*\*before\*\*** the next plaintext is inspected.

---

## ## 27. Verification-script status

The repository currently contains dedicated verification scripts for Volumes 1 and 2.

Volume 3 does not yet have a dedicated verifier.

This Markdown edition therefore keeps the key coordinates, transforms, phase arithmetic, and mod-29 subtraction explicit so that a future `verify\_volume\_3.py` can test the same claims directly.

Related repository files:

- [Volume 2 technical Markdown](./0-2-volume-2.md)
- [Raw 0-2 rune dataset](../0-2-runes.txt)
- [27×27 grid image](../grid.png)
- [Verification scripts](../verify/)
- [Repository README](../README.md)

---

## ## 28. Current strongest plaintext

The current connected plaintext candidate is:

> **AS I GO, THE WEATHER TURNS COLD. I MAY CRY NOW. THE ...**

The new Volume 3 contribution is:

> **NOW THE**

This continuation is strongest when treated as part of the same sequential route rather than as an isolated crib.

---

## # 29. Research conclusion

Volume 3 adds a geometric layer to the model developed in Volumes 1 and 2.

The key new proposal is not only that mirror structures are symmetric, but that their symmetry can carry **directional information**:

```text  
outer rune → toward center
```

At the `CRY` endpoint, the inherited value `6` is unusually constrained:

```text  
E-G-E signature
↓
6
↓
previous movement
↓
radius of S mirror
↓
 ϕ^2 of IA/O center
↓
radius of TH mirror
↓
U6 transition
↓
TH(11,16)
```

The subsequent key is generated by an already established rule, its phase is fixed numerically, and the five-rune ciphertext decrypts to:

> \*\*NOW THE\*\*

The broader working hypothesis is therefore that \*\*totient arithmetic and mirror geometry are not separate observations but interacting layers of the same transition system\*\*.

The next major test is predictive:

> Can the state after `NOW THE` determine the next movement, structure, key, and plaintext before the continuation is known?

---

## 30. Minimal machine-readable chain

```text

VOLUME_2_END:

plaintext = "AS I GO, THE WEATHER TURNS COLD. I MAY CRY"

cry_ciphertext = J-OE-S

cry_key = G-E-E

cry_endpoint = S(17,16)

PREVIOUS_NODE:

E-X-E

COMPILE:

$\phi(X=14)=6=G$

E-X-E $\rightarrow$ E-G-E

SIGNATURE:

E-G-E $\rightarrow$ 6-2-6

inherited_value = 6

CRY_ENDPOINT_MIRROR:

S(17,4)

IA/O(17,10)

S(17,16)

radius = 6

CENTER_CHECK:

IA/O = 27

$\phi(27) = 18$

$\phi(18) = 6$

$\phi^2(IA/O) = 6$

COMPANION_MIRROR:

TH(11,16)

IA/O(17,10)

TH(23,4)

radius = 6

MOVE:

S(17,16)

U6 $\rightarrow$ TH(11,16)

CIPHERTEXT_COORDINATES:

TH(11,16)

AE(12,17)
B(13,18)
A(14,19)
J(15,20)

CIPHERTEXT:
TH-AE-B-A-J

KEY_NODE:
OE-J-OE

KEY_COMPILE:
J=11
 $\phi(11)=10=I$
OE-J-OE $\rightarrow$ OE-I-OE

KEY_SIGNATURE:
OE-I-OE $\rightarrow$ 10-4-10

PHASE:
 $\mu(10)+\mu(4)+\mu(10)$
 $= 1+0+1$
 $= 2 \bmod 3$
phase = 2

ACTIVE_KEY:
OE-OE-I
repeated = OE-OE-I-OE-OE

DECRYPT:
 $TH(2)-OE(22) \bmod 29 = 9=N$
 $AE(25)-OE(22) = 3=O$
 $B(17)-I(10) = 7=W$
 $A(24)-OE(22) = 2=TH$
 $J(11)-OE(22) \bmod 29 = 18=E$

PLAINTEXT:
N-O-W-TH-E
"NOW THE"

CURRENT_RESULT:
"AS I GO, THE WEATHER TURNS COLD. I MAY CRY NOW. THE ..."

\\

> **Research status:** proposed, reproducible in the documented arithmetic and coordinates, and still under active analysis.

> This document does not claim an officially verified Cicada 3301 solution.